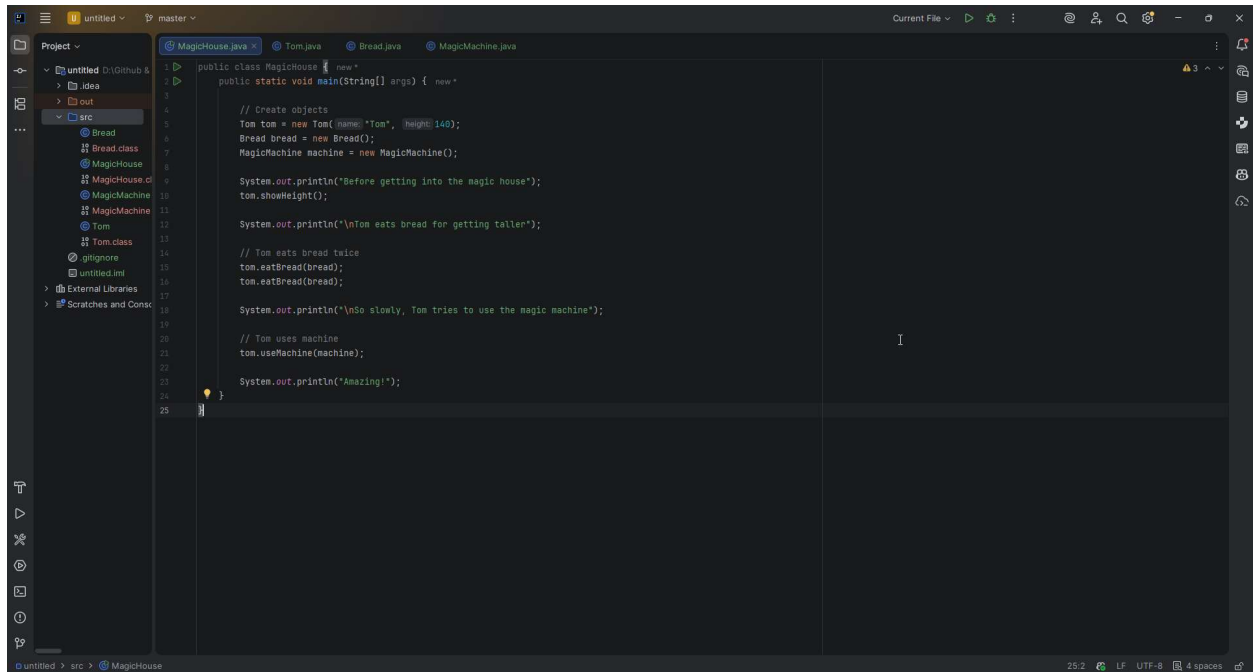


Assignment 4

Student Number: F24040229

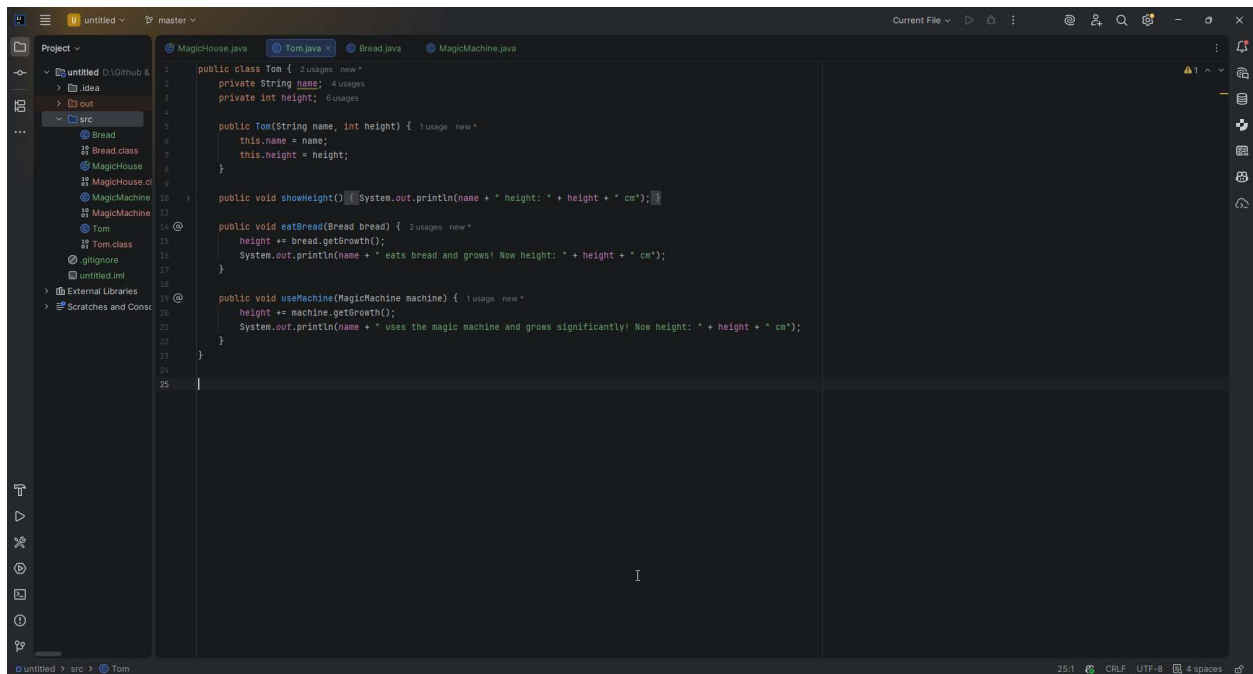
Student Name: Ahmod Abdullah

My Work:



This screenshot shows the MagicHouse.java file in the MagicHouse package. The code defines a main method that creates objects for Tom, Bread, and MagicMachine, and then simulates Tom's actions of getting into the house, eating bread, and using the magic machine.

```
1 public class MagicHouse {  
2     public static void main(String[] args) {  
3         // Create objects  
4         Tom tom = new Tom("Tom", 140);  
5         Bread bread = new Bread();  
6         MagicMachine machine = new MagicMachine();  
7  
8         System.out.println("Before getting into the magic house");  
9         tom.showHeight();  
10  
11         System.out.println("\nTom eats bread for getting taller");  
12  
13         // Tom eats bread twice  
14         tom.eatBread(bread);  
15         tom.eatBread(bread);  
16  
17         System.out.println("\nSo slowly, Tom tries to use the magic machine");  
18  
19         // Tom uses machine  
20         tom.useMachine(machine);  
21  
22         System.out.println("Amazing!");  
23     }  
24 }  
25 }
```



This screenshot shows the Tom.java file in the Tom package. The code defines the Tom class with attributes name and height, and methods for showing height, eating bread, and using the magic machine.

```
1 public class Tom {  
2     private String name;  
3     private int height;  
4  
5     public Tom(String name, int height) {  
6         this.name = name;  
7         this.height = height;  
8     }  
9  
10    public void showHeight() {  
11        System.out.println(name + " height: " + height + " cm");  
12    }  
13  
14    public void eatBread(Bread bread) {  
15        height += bread.getGrowth();  
16        System.out.println(name + " eats bread and grows! Now height: " + height + " cm");  
17    }  
18  
19    public void useMachine(MagicMachine machine) {  
20        height += machine.getGrowth();  
21        System.out.println(name + " uses the magic machine and grows significantly! Now height: " + height + " cm");  
22    }  
23 }  
24  
25 }
```

This screenshot shows an IDE window with the file `Bread.java` open. The project structure on the left includes `src` with `Bread`, `Bread.class`, `MagicHouse`, `MagicHouse.cl`, `MagicMachine`, `MagicMachine`, `Tom`, `Tom.class`, `gitignore`, `untitled.txt`, `External Libraries`, and `Scratches and Cons`. The code in `Bread.java` is as follows:

```
1 public class Bread { 3 usages new *
2     private static final int GROWTH = 1; // 1 cm per bread 1 usage
3
4     public int getGrowth() { 1 usage new *
5         return GROWTH;
6     }
7
8 }
9
```

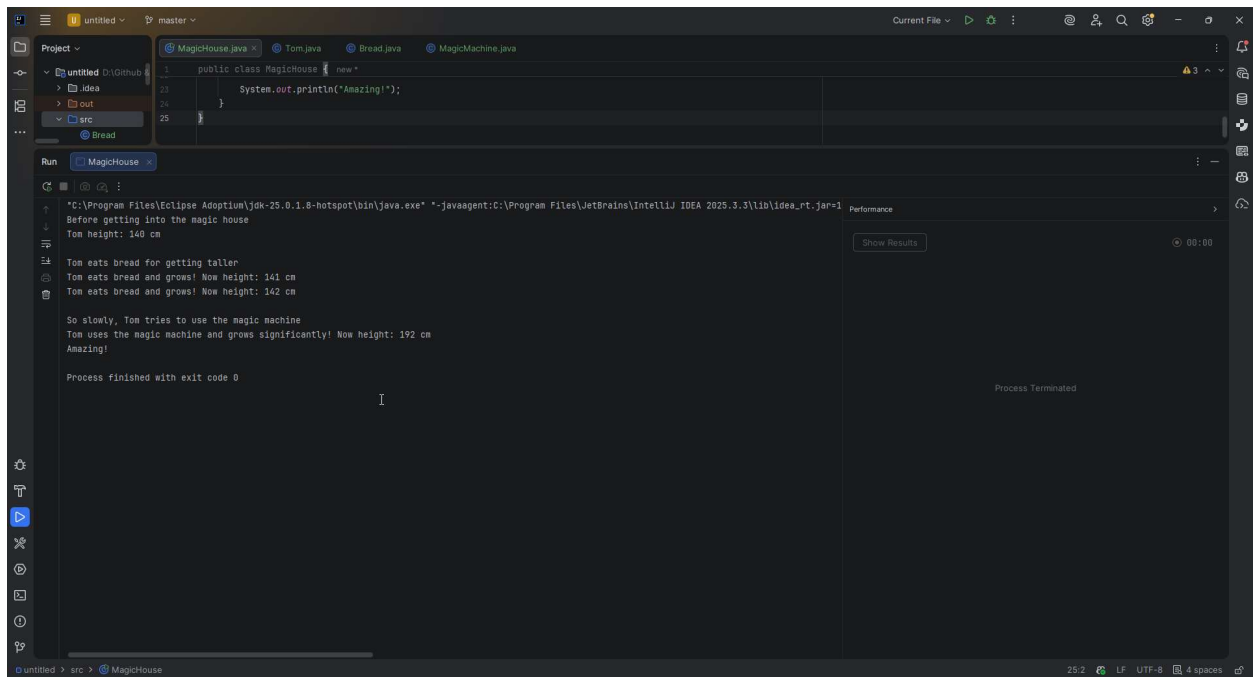
The status bar at the bottom indicates the file encoding is UTF-8 with 4 spaces.

This screenshot shows the same IDE window with the file `MagicMachine.java` open. The project structure on the left is identical to the previous screenshot. The code in `MagicMachine.java` is as follows:

```
1 public class MagicMachine { 3 usages new *
2     private static final int GROWTH = 50; // 50 cm growth at a time 1 usage
3
4     public int getGrowth() { 1 usage new *
5         return GROWTH;
6     }
7
8 }
9
```

The status bar at the bottom indicates the file encoding is UTF-8 with 4 spaces.

Output:



```
Run MagicHouse
"C:\Program Files\Eclipse Adoptium\jdk-25.0.1.8-hotspot\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\lib\idea_rt.jar=1
Before getting into the magic house
Tom height: 140 cm

Tom eats bread for getting taller
Tom eats bread and grows! Now height: 141 cm
Tom eats bread and grows! Now height: 142 cm

So slowly, Tom tries to use the magic machine
Tom uses the magic machine and grows significantly! Now height: 192 cm
Amazing!

Process finished with exit code 0
```

Performance

Show Results 00:00

Process Terminated